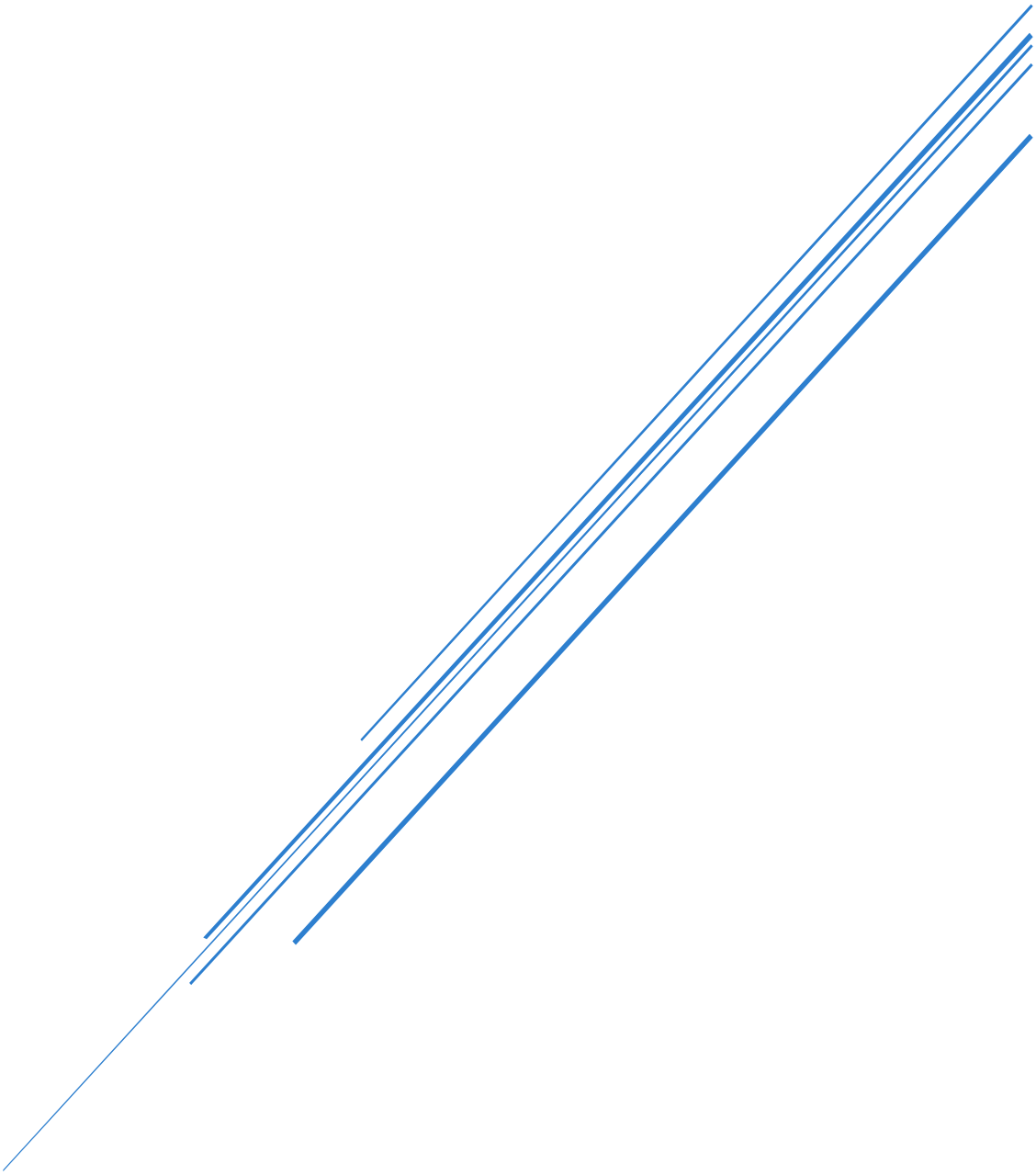
MINI PROJECT 1:

Northwest Architecture Studio Ltd



By Aiman, Seerat, Jimmy and Faris

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Introduction

As part of a small computing team advising Northwest Architecture Studio Ltd, we are helping Northwest in a project where we are required the replacement of 12 computer workstations across 4 separate roles of employees.

The company does not want to buy the same PC for every staff member because separate roles have different hardware requirements and budget allocations, requiring workstation recommendations that are suitable, affordable, and capable of supporting the employees' daily tasks.

This project only includes hardware and does not include Operating System Licences, Office Software (OS), CAD software, antivirus subscription, and any other software licenses.

The project includes the workstations of the following roles:

- Administrative Assistant (3 workstations)
- Project manager (2 workstations)
- Architect / CAD Designer (5 workstations)
- 3D Visualisation Specialist (2 workstations)

This report contains:

Role Description of every employee role and why this might affect the workstations hardware requirements.

Hardware Specification: including what components are bring used and all the required peripherals.

Buildcores Screenshots: Evidence to justify what the setup should looks like as well as the prices evidence.

Cost table: Showing the price evidence and calculations of every component and total role cost.

Component Justification: explaining all the choices.

Administration Assistant:

1. Role description

The administrative assistant keeps the office running smoothly by managing day-to-day clerical, organization, and communications task. Their responsibilities include answering phone calls, attending online meeting, and replying to emails. One of their most important responsibilities is recording and storing data, such as information about business partners and customers. Because some of this data is confidential, it must be handled securely. their computer literacy is basic to intermediate, so they need a workstation that is simple to work and reliable to use.

The nature of this role has a direct impact on the hardware it needs. Administrative assistant does not need a powerful computer, because their work does not involve demanding task. The programs they use, such as excel and PowerPoint, are basic office software that almost any modern computer can run smoothly.

However, the most demanding task is video calls. Especially if outlook and a web browser are open at the same time. As result the workstation need enough RAM to avoid slowdown, a stable internet connection, and webcam headset.

Overall, this role needs a balanced, affordable computer, not a powerful one. The assistant admin is not going to do any programming, so any overpowered PC would be a waste of money. It is better spent on other departments, such as CAD designer who needs a powerful station.

2. Hardware requirements

- **CPU:** a mid-range processor with integrated graphics, so a separate graphics card is not needed. This saves money on parts and reduces electricity use.

- **RAM:** 16GB, because the assistant often has a web browser, outlook, and other programs open while running video calls at the same time. Less than 16GB could cause the office programs to slow down. The integrated graphics also use part of the RAM, leaving less memory for other programs.

- **Storage:** a 512GB SSD, so the assistant can work on their files smoothly. 512GB is enough space for windows, office programs, and documents, because these files are small, unlike 3D models.

- **GPU:** not required for the daily tasks of an administrative assistant. Graphics cards are needed for roles such as CAD and 3D visualisation**,** because they work with complex 3D models with a lot of details.

- **Monitor:** 24-inch full HD screen, which is big and clear enough for daily tasks such as spreadsheet and emails. features like Matte screen and Low Blue mode support the wellbeing of someone who sits at a desk for 8hours or more.

- **Keyboard/ mouse**: a wired keyboard and mouse is ideal for an office-based job, because there no batteries to run, it out also have number pads, which it helps type in numerical data quickly, for example in excel.

- **Webcam and headsets:** needed for online meetings with colleagues and potential with future clients. A 720p webcam is good enough for the calls. The headset's noise-cancelling microphone can improve the quality of a conversation. this is needed in a busy office.

- **Network:** A wired Ethernet network connection is the most stable option for office-based role. this is important for video calls, shared calendars, and instant messaging through Microsoft teams. Having a built-in Wi-Fi also provides a backup, this is called redundancy. This is important if one connection fails.

- **PSU and case:** A 550W power supply is enough because the parts only use 159W and the role does not need high performance. This leaves a good safety margin and keeps the system quiet and efficient. With a simple case (build for office) with front USB ports and audio jacks, make it easy for staff to plug in devices. Spending more money on a gaming design.

3. Build specification

The base unit for this set up was configure on buildcores and the peripherals were sourced from UK retailers.

Component	**product**	**Retailer**	**cost**
CPU	AMD Ryzen 5 5600G (Coolers include)	LaptopOutlet	£154.99
Motherboard	Asus prime B550m-A Wi-Fi II AM4 DDR4 mATX	CCL	£87.99

RAM	Corsair vengeance LPX (2x8GB) DDR4 3200 CL16 Black	Overlockers	£149.99
GPU	None- Integrated graphics		£0.00
Storage	Patriot P300 512 GB SSD M.2-2280 PCIe X4 NVMe	CCL	£68.99
PSU	MSI A550BN Black 550W 80+ Bronze Certified	BOX	£40.60
Case	KOLINK KLA-003 ATX Mid Tower Black	Overlockers	£17.99
Cooling	Included with CPU		£0.00
monitor	Philips 24E1N1100-24-inch Full HD monitor, 120 Hz, Adaptive Sync. (1920x1080, 1x D-Sub, 1x HDMI 1.4) black	Amazon	£68.60
headset	Jabra Evolve 10 wired USB Headset with Noise – Cancelling Mic	Amazon	£ 23.95
webcam	Logitech C270 720p 1280 x 720 pixels; 1280 x 720pixels USB 2.0 Webcam – Black Webcams (3 MP, Clip, USB 2.0, Black)	Amazon	£19.89
keyboard/mouse	TECKNET Wired Keyboard and Mouse Set, Full-Size, UK QWERTY Layout, Black	Amazon	£18.99
Network cable	UGREEN Ethernet Cable 5M, Flat RJ45 LAN	Amazon	£5.05

Cost	**Amount**
Base unit subtotal	£520.55
Peripherals Subtotal	£136.48
Delivery charges	£0.00
Total per workstation	£657.03
Total for 3 workstations	£1,971.09

4. Budget calculations

	Per workstations	**Total (3 workstation)**
Hardware cost	£657.03	£1,971.09
Budget	£950.00	£2,850.00
Under budget by	£292.97	£878.91

% of budget used	69.2%	69.2%

The total cost per workstation is £657.03 and the total for all three workstation is £1,971.09. this shows that the build is within budget, as only 69.2% of the £2,850.00 role budget has been used, leaving £878.91 unspent

5. Supporting screenshots

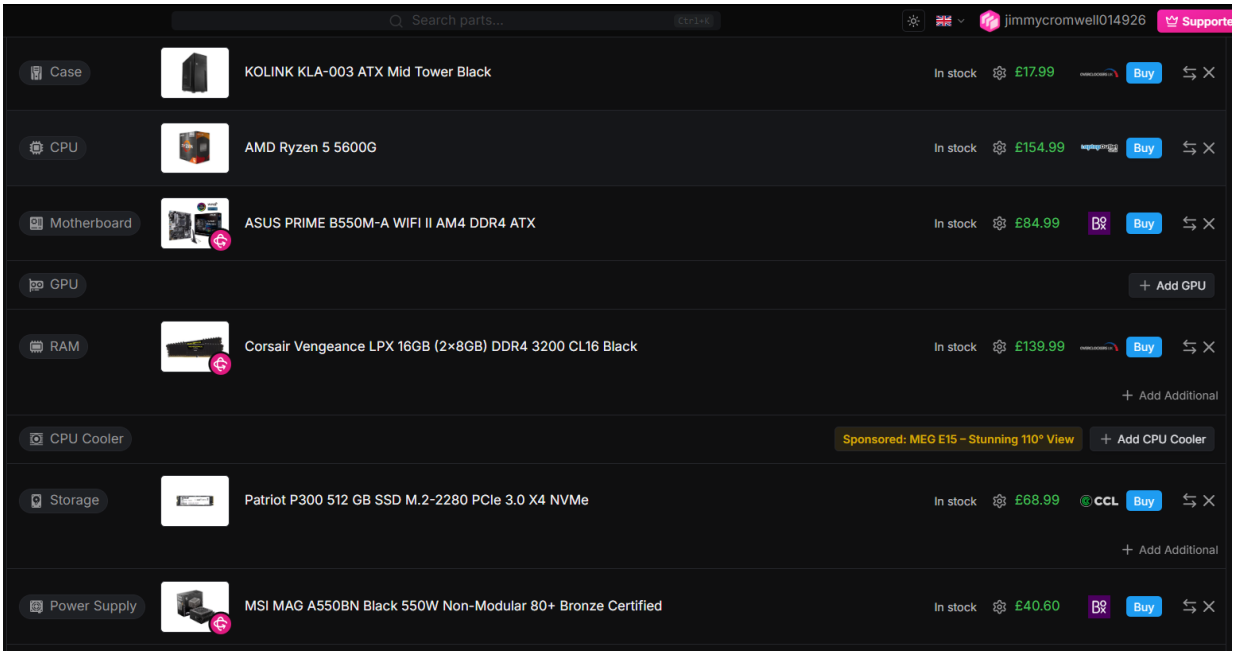


Figure 1: base unit specification on buildcores (checked 02/10/2026).
Buildcores listed the ram price as £139.99, but the retailer price checked on overlockers was £149.99, so the verified retailer price was used.



Figure 2: AMD Ryzen 5 5600g on laptopoutlet (£154.99 checked on 02/10/2026)



Figure 3: ASUS Prime B550M-A WIFI II Motherboard on CCL (£87.99, checked 02/10/2026)



Figure 4: Corsair vengeance LPX (2x8GB) DDR4 3200 CL16 Black (£87.99 checked 02/10/2026)



figure 5: Patriot P300 512 GB SSD M.2-2280 PCIe X4 NVMe (£68.99 Checked 02/10/2026)

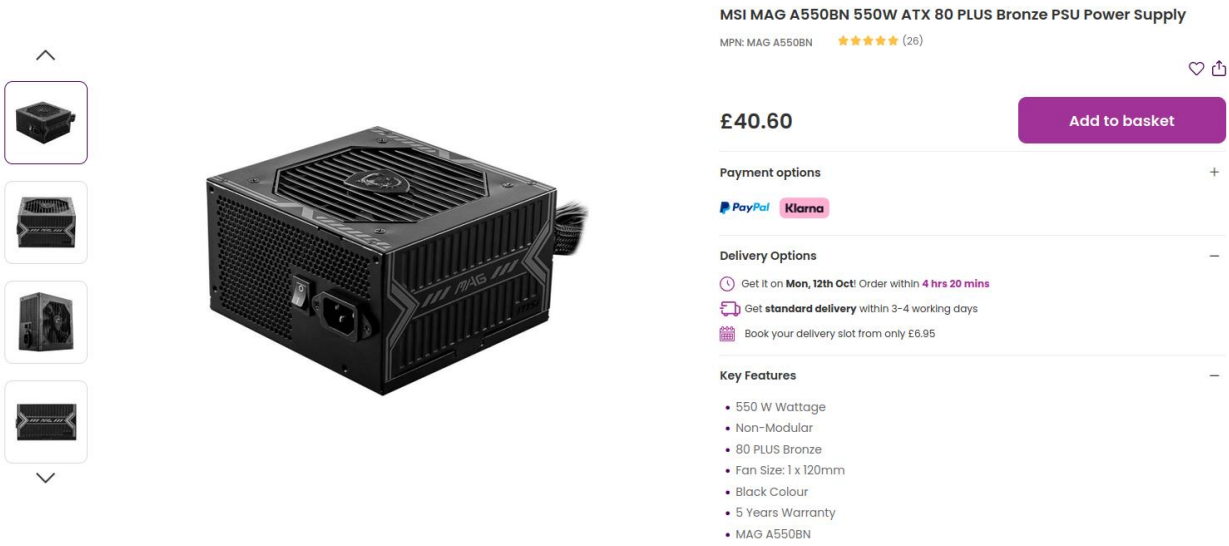


Figure 6: MSI A550BN Black 550W 80+ Bronze Certified on Box (£40.60 Checked on 2/10/2026)

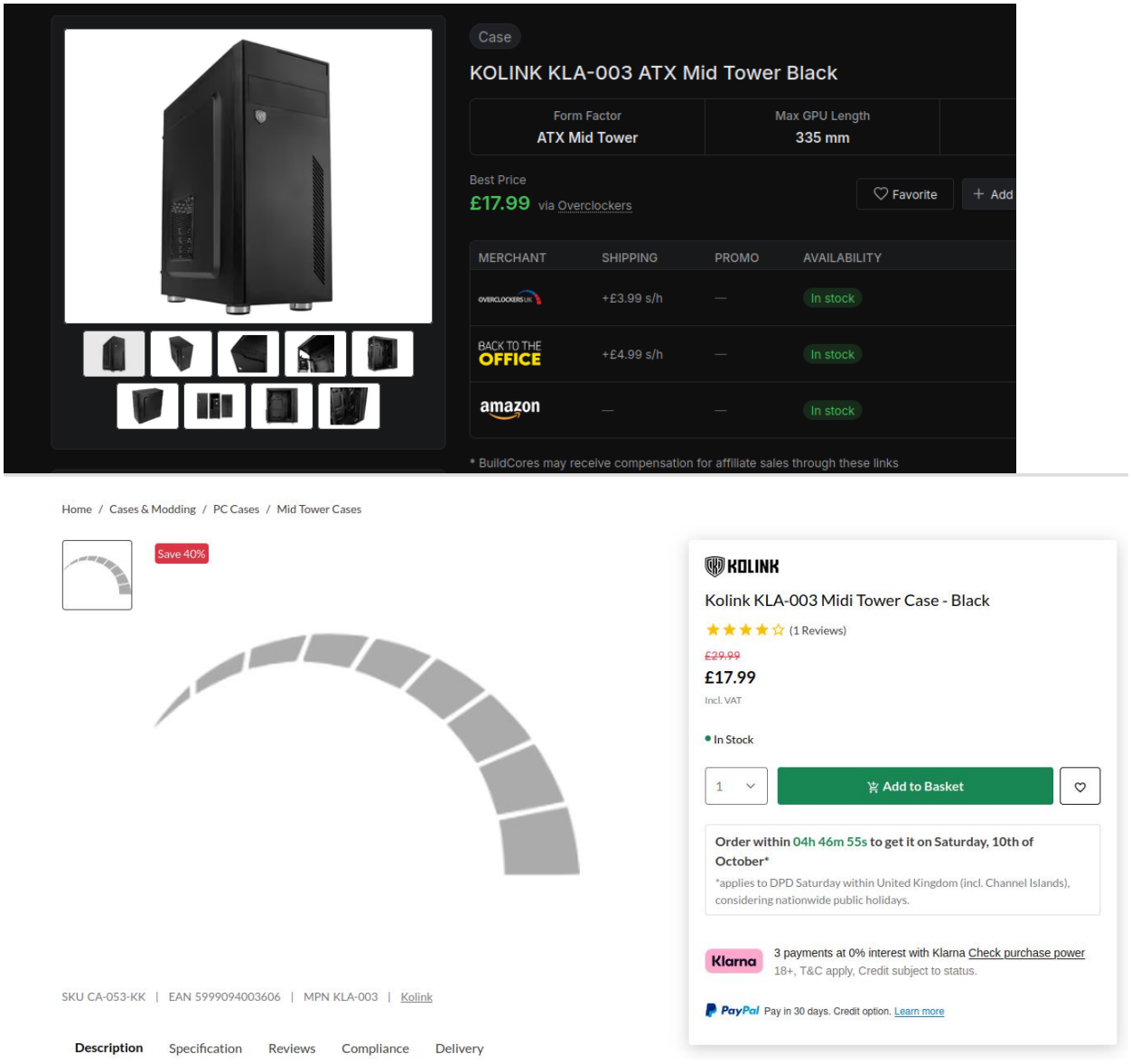


Figure 7: KOLINK KLA-003 ATX Mid Tower Black case (overlockers £17.99 checked on 02/10/2026



Figure 8: Philips 24E1N1100-24-inch Full HD monitor, 120 Hz, Adaptive Sync. (1920x1080, 1x D-Sub, 1x HDMI 1.4) black on Amazon (£68.60 checked on the 04/10/2026)



Figure 9: headset Jabra Evolve 10 wired USB Headset with noise-cancelling Mic. Amazon (£23.36 checked on the 4/10/2026)



Figure 10: webcam Logitech C270 720p 1280 x 720 pixels; 1280 x 720pixels USB 2.0 Webcam – Black Webcams on amazon (3 MP, Clip, USB 2.0, Black)



Figure 11: TECKNET Wired Keyboard and Mouse Set, Full-Size, UK QWERTY Layout, Black (£18.99 on amazon checked on 4/10/2026)



Figure 12: Network cable UGREEN Ethernet Cable 5M, Flat RJ45 La (Amazon, £5.05 checked on the 4/10/2026)

6. Component justification

CPU AMD RYZEN 5 5600G: the administrative assistant uses, word, excel and teams at the same time, and the 5600G's 6 cores and 12 threads allow it to run them without slowing down. Its integrated graphics "G" removes the need for a separate graphics card. Choosing the RYZEN 5 5600 (£119.99, Amazon) checked on Amazon on 3/10/2026) would have meant adding a graphic card such as the GT 710 (£54.99 checked on ccl on 3/10/2026), costing £19.99 more per workstation.

A more powerful RYZEN like a RYZEN 7 5800X (£189.99 checked on Amazon on the 3/10/2026) would also need a graphics' card, costing £89.99 more per workstation. Its extra power is not used for office tasks, so it would waste company funds. The 5600G also includes a cooler, which avoids a further cost. The money saved can instead be used and support CAD and 3D visualisation roles, which need more powerful hardware.

RAM CORSAIR VENGEANCE LPX 16GB(2x8GB) DDR4 3200 CL16

The administrative Assistant regular runs word, excel, PowerPoint, teams, and a web browser with multiple tabs at the same. Microsoft list 4GB of ram as the minimum for teams and recommends 8GB for its advanced feature (Microsoft ,2026a). however, this is teams alone. with only 8GB, there would not be enough memory for all these programs, causing the computer to slow down. The integrated graphics cards. also use part of the system RAM. So 16GB is the sweet spot for communication, scheduling, and organisation tools.

A 32GB kit (£239.99, amazon) would cost £90.00 more per workstation and is better suited for to designs roles (3D visualization). for this role it would be over -specifying and a waste of company funds.

Two 8GBS sticks whereas DDR4 were chosen instead of one 16GB stick to enable dual channel mode. with singles stick, data travelling between RAM and the CPU can become congested, especially when several programs are sending data at the same time, where does two sticks allow more data to be transferred at once. this is even more important because the 5600G's integrated graphics have no memory of their own and share the system RAM with the CPU. The DDR4 was chosen because the motherboard only supports DDR.

Storage

Storage An NVMe SSD was chosen instead of hard drive (HDD), because an HDD is much slower and would cause delays when starting the computer and opening files. And HDD also has moving parts, which makes it more likely to fail.
The 512GB Patriot P300 (£68.99) cost only £7.00 more than the 256GB version (£61.99, CCL) but offers double the space. the 1TB version is (£132.99, amazon) would cost £64.00 more per workstation. This extra space is not need, because the administrative Assistant works with small office file sand customers siles should be saved on the company server or cloud with backups rather than only on the computer. Choosing 1TB would waste company funds.

GPU

No dedicated Graphics cards included, because the administrative assistant only carries out light tasks such as word, excel, teams and replying to emails in outlook. These tasks only need one monitor, and the integrated graphics of the Ryzen 5 5600g are more than enough for this. Adding a graphics card such as the GT 710 (£54.99), would increase the cost without improving performance for this role.

Monitor

The Philips 24EN100A (£68.60, Amazon was chosen because it has full HD (1080p) resolution, which is enough for word, excel. Outlook and teams. A 4k monitor is better suited to the CAD

and 3D visualisation roles, which need excellent details. A 24-inch screen was chosen instead of a 27-inch one, such as the GAMEON 27 FHD (£74.99, amazon), which would cost £6.30 more per workstation (£19.17 across three). At the same time resolution, a large screen has lower pixel density, which makes text less sharp and at desk distance it requires more neck and eye movement.

The monitor uses IPS panel, which gives accurate colours and wide view angles, so the image does not change colour when a colleague views it from the side during a video call. The Low Blue mode reduces blue light emissions, which lessens eye fatigue during a full working day. The monitor concepts to the motherboard's HDMI port, as there is no response graphics card.

Peripherals

A wired TECKNET keyboard and mouse (£18.99, Amazon) was chosen instead of a wireless set (£19.99, amazon). the price different was only £1.00 per workstation. but a wired set is more reliable there are not batteries that can run out and no connection that can drop while the administrative Assistant is working on important documents.

The Jabra evolve 10 headset (£23.95, amazon) is used for teams calls and meetings. Its microphone allows the person on the other end to hear clearly, even in a busy office. it also keeps calls private, so colleagues cannot overhear sensitive customer data, which could happen with speakers. It is wired for the same reasons as the keyboard.

The Logitech C270 (£19.89, Amazon) webcam records 720p. teams can deliver up to 1080p video, but only when bandwidth is not limited. When bandwidth is insufficient, its priorities audio over video and the quality also depends on the number of people in a meeting (Microsoft ,2026b). A 1080p webcam like the Logitech C920s (£42.49, Amazon) would therefore not always give a real benefit, while costing £22.60 more per workstation (£67.80 across three)

A 5m UGREEN cat6 cable (£5.05, amazon gives a reliable, wired connection for video calls, while the motherboard's Wi-Fi can act as backup. The 5m length allows the desk to be placed some distance from the wall socket.

Motherboard, PSU, and case:

The ASUS Prime B550M-A WIFI II (£87.99, CCL) was chosen because the Ryzen 5 5600G uses the AM4 socket, so it can only fit an AM4 motherboard. The board also provides the M.2 slot for the SSD, the HDMI port for the monitor and WI-FI as a backup connection.

The MSI MAG A550BN (£40.60) provides 550W while the build uses about 159W.A PSU that can run at full power can overheat and stop working very easily. so that the extra headroom allows it to run safely. its 80+bronce rating means that around 82%-85% of the power taken from the wall reaches the computer, with less energy wasted as heat.

The Kolink KLA-003 (£17.99, overlockers) was chosen because all the components fit inside it. the first option, the Kolink KLM-002, only supports SFX power supplies, so the ATX PSU would not have fitted.

7. Compromises

Integrated graphics: Without a dedicated graphics card, the workstation cannot handle demanding tasks such as modern video games or 3D rendering; However, it is well suited to the office tasks the administrative Assistant carries out. if more graphics power is needed in the future, a graphics card could be added, as the 550W PSU provides plenty of headroom above the builds 159W.

512GB storage: if the SSD becomes full, new files cannot be saved. so, the system performance drops and windows updates may fail. this risk is reduced because customers files are stored on the company server. A second drive can be added without replacing the existing one.

720p Webcam: the videos images are less sharp than a 1080p webcam. It still has HD quality, this is good enough for team meetings, as the webcam connects by USB, it can easily be replaced if higher quality is needed in the future.

8. Ethical and responsible purchasing

Energy use: The workstation uses about 159W, as it has no dedicated graphics card. The 80+ Bronze PSU waste only a small amount of energy as heat, and the Patriot P300 SSD has a low-power design. this reduces both energy costs and environmental impact across three workstations.

Less waste: the wired keyboard and the mouse do not need batteries, because they are not wireless. So, there is no need to buy new batteries or throw old ones. This reduces waste management in the business.

Employee wellbeing: The monitor's Low Blue mode helps reduce eye fatigue and its IPS panels gives clear colour from different angles, which support the administrative Assistant during a full working day**.**

Data protection: The headset keeps calls private, so if there is any sensitive data that it has been shared cannot be heard. customers files should be stores on company sever with backups so that their data is not lost if the SSD fails.

Responsible use of budget: No unnecessary components were included which leaves more of the budget for the CAD and 3D visualisation roles that need more powerful hardware for their roles.

Project manager:

1. Role Description:

As the project manager organizes plans and makes sure that deadlines and budgets and tasks are managed effectively. I use spreadsheets, reports, emails, and project management software to communicate with clients and team members I also use the software to organize work etc. I regularly attend video meetings, and I need to have several different applications up and running at the same time. So that is why I need a responsive and fast workstation with enough rams to multitask fast storage for applications and files and two monitors' I can view timelines documents and project information side to side. A good headset and webcam are required for professional and clear communication during online meetings, etc.

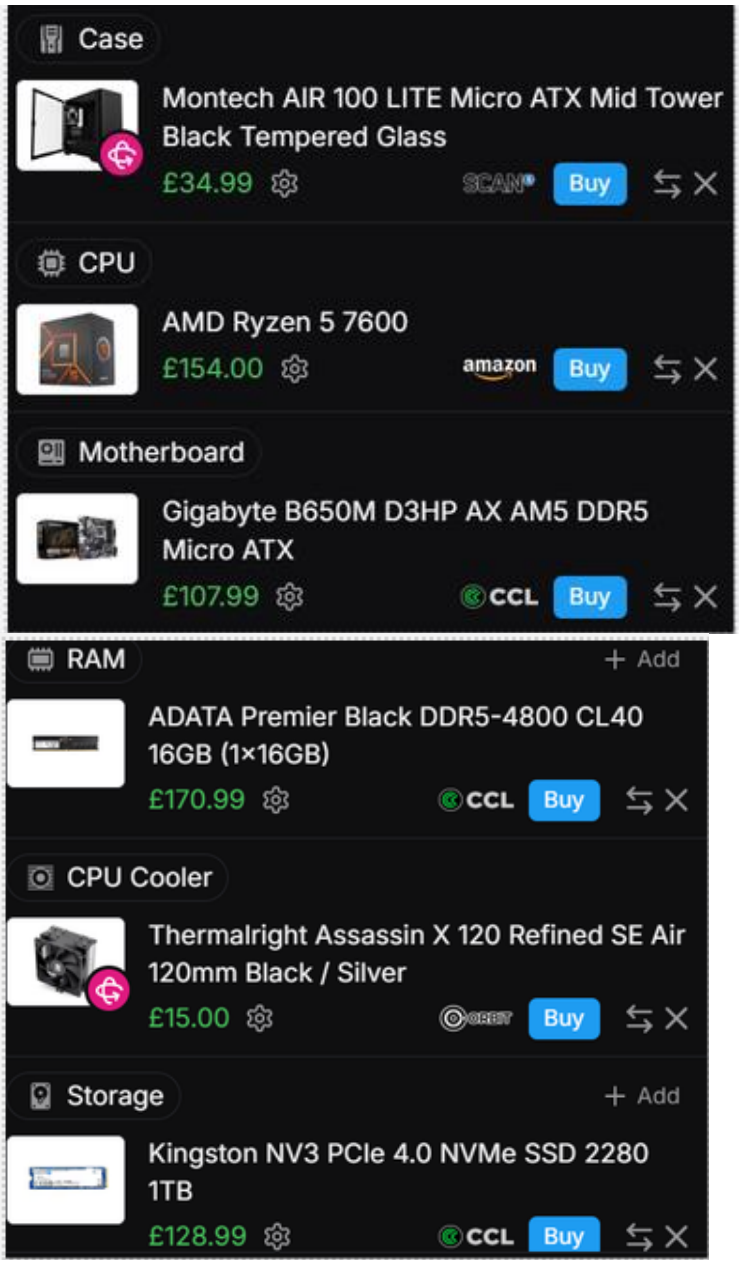
2. Hardware Specification

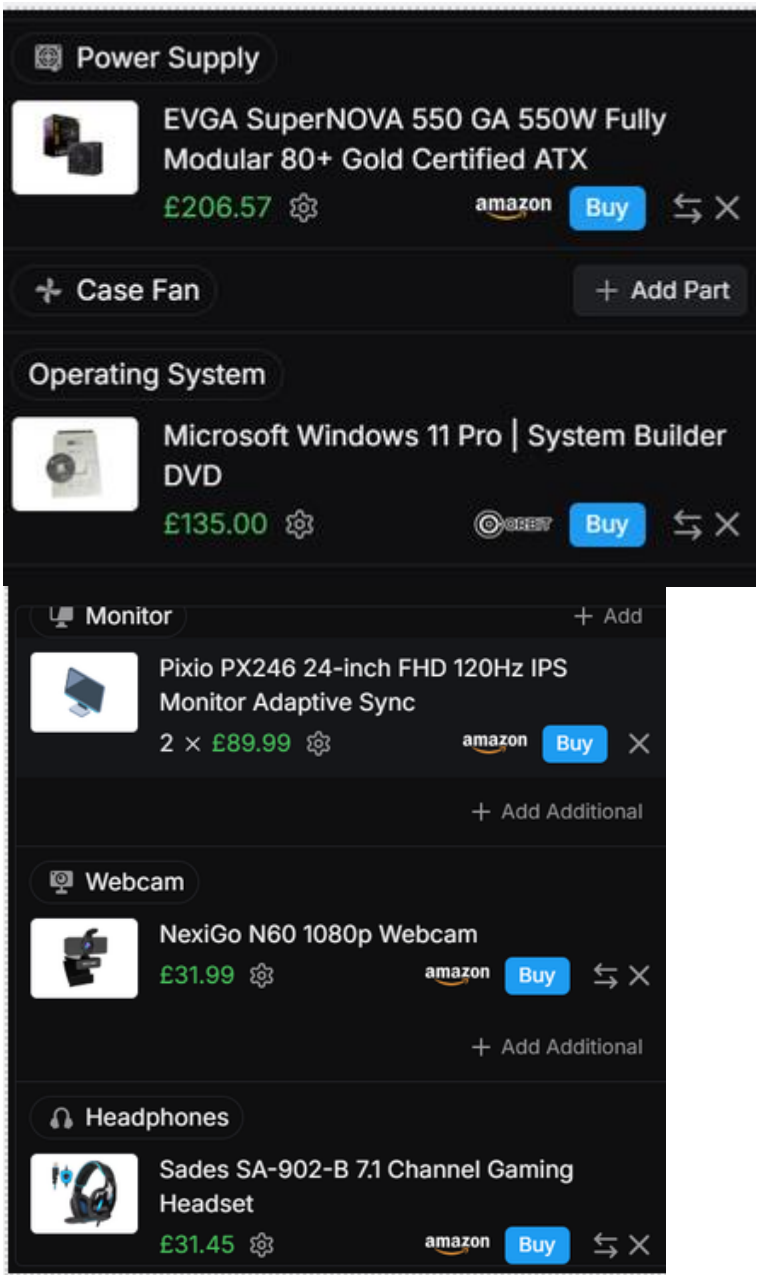
Component	Specification	Qty.	Price Each	Purpose
CPU	AMD Ryzen 5 7600	2	£154.00	Fast multitasking
Motherboard	Gigabyte B650M D3HP AX	2	£107.99	Compatibility & connectivity
RAM	16GB DDR5 (2×8GB)	2	£170.99	Multitasking
Storage	Kingston NV3 PCIe 4.0 NVMe 1TB	2	£128.99	Fast loading & file storage
GPU	none	2	£0.00	Dual-display office workflow
CPU Cooler	Thermalright Assassin X 120 Refined SE	2	£15.00	CPU cooling
PSU	EVGA SuperNOVA 550 GA 550W Fully Modular 80+ Gold	2	£206.57	Reliable power
Case	Montech AIR 100 Lite	2	£34.99	Airflow & protection
Operating System	Windows 11 pro	2	£135.00	Operating system for running business applications
Monitors	Pixio PX246 24-inch FHD 120Hz IPS	4	£89.99	Dual-monitor workflow
Keyboard + Mouse	Dell KM5221W keyboard & mouse	2	£29.99	Daily input
Webcam	NexiGo N60 1080p	2	£31.99	Client meetings
Headset	Sades SA-902-B 7.1 Channel	2	£31.45	Communication

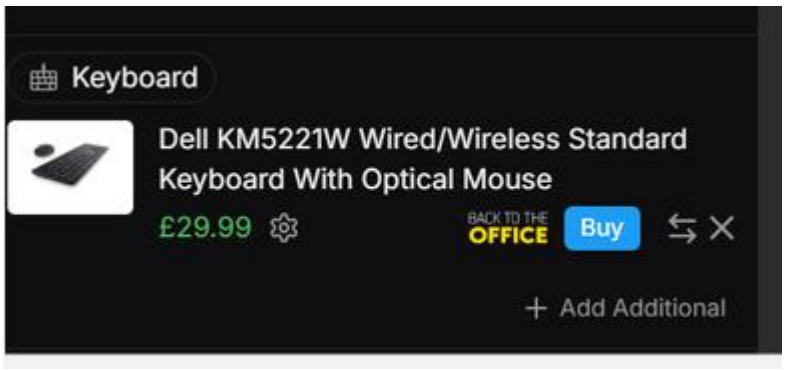
3. BuildCores Build Screenshot



4. Pricing Evidence







5. Item Cost Table

Item	Cost Each	Qty.	Total Cost	Justification
CPU – AMD Ryzen 5 7600	£154.00	2	£308.00	Fast multitasking

Motherboard – Gigabyte B650M D3HP AX	£107.99	2	£215.98	Compatibility
RAM – ADATA Premier Black 16GB DDR5	£170.99	2	£341.98	Multitasking
SSD – Kingston NV3 1TB PCIe 4.0 NVMe	£128.99	2	£257.98	Fast storage
GPU – None / integrated graphics	£0.00	2	£0.00	Display support
CPU Cooler – Thermalright Assassin X 120 Refined SE	£15.00	2	£30.00	CPU cooling
PSU – EVGA SuperNOVA 550 GA 550W Gold	£206.57	2	£413.14	Reliable power
Case – Montech AIR 100 Lite	£34.99	2	£69.98	Cooling
Windows 11 Pro	£135.00	2	£270.00	Business OS
Monitors – Pixio PX246 24" FHD 120Hz IPS	£89.99	4	£359.96	Dual-screen workflow
Keyboard + Mouse – Dell KM5221W	£29.99	2	£59.98	Input
Webcam – NexiGo N60 1080p	£31.99	2	£63.98	Meetings
Headset – Sades SA-902-B 7.1 Channel	£31.45	2	£62.90	Communication
TOTAL – TWO WORKSTATIONS			**£2,453.88**	
TOTAL – PER WORKSTATION			**£1,226.94**	**Under £1,300 maximum**

Budget result: UNDER budget (Saves £73.06 per workstation / £146.12 total for the role).

6. Component Justification

CPU AMD Ryzen 5 7600

I chose the AMD Ryzen 5 7600 due to its high performance with its 6 cores and 12 threads, this makes it extremely good at running office software such as excel and spreadsheets, the 6 cores in it allow it to multitask seamlessly, with large spreadsheets and emails and several other things it can run without being the bottleneck. It also comes to work Radeon graphics, which is useful as this means dedicated graphics is not needed.

Motherboard Gigabyte B650M D3HP AX

This motherboard holds the compatible CPU slot or the Ryzen 5 7600, it also supports high speed connections such as having slots compatible with DDR5 RAM, it also comes with built in Wi-Fi. Display connections and Bluetooth, making this

suitable for its high connectivity and compatibility paired with speed. It also allows a dual monitor system which would help greatly when multitasking.

RAM ADATA Premier Black DDR5

I chose DDR5 RAM ad this provides the highest speeds available with very little latency, as well as being able to handle multiple applicator for the project manager, such as dashboards, spreadsheets, and various applications running at the same time, the higher the capacity the higher the load the workstation can handle, DDR5 is extremely fast and will run the applications smoothly, and would reduce the slowdown when several programs are being used at once.

Storage Kingston NV3 1TB PCIe 4.0 NVMe SSD:

I chose the Kingston NV3 because it provides incredibly fast storage for Windows, applications, and project files, NVME drive is much faster than ant SSD and hard drive by a lot, and this would reduce the workstation booting time to fractions off what any hard drive or SSD can give. The 1TB capacity will give the Project manager enough space for work related files and will maintain a fast response time. Kingston lists the 1TB model as a PCIe 4.0 NVMe SSD with speeds of up to 6,000MB/s read and 4,000MB/s write.

GPU, Integrated Radeon Graphics:

I chose an integrated graphics card because the Project Manager uses the workstation for office, communication, and project management tasks with CPU intensive and does not need a dedicated graphics card, as a 3d application would have needed. The Ryzen 5 7600 already includes Radeon graphics, so a separate GPU is unnecessary for this role.

CPU Cooler, Thermalright Assassin X 120 Refined SE:

I chose this CPU cooler to provide dedicated cooling for the processor during extended periods of multitasking. As the CPU under heavy load abs multitasking will slow down due to elevated temperature and would potentially lower its lifespan, good cooling helps maintain stable performance and prevents excessive heat during extended working sessions.

PSU, EVGA SuperNOVA 550 GA 550W 80+ Gold:

I chose the 550W EVGA SuperNOVA power supply because it provides reliable and sufficient power for the workstation components. The 80+ Gold efficiency rating also makes it appropriate for a business workstation that may be running many hours each day; this PSU would be the minimum standard for such tasks.

Case, Montech AIR 100 Lite:

I chose Montech AIR 100 Lite because it provides airflow and protection for the internal components. Good airflow is important for keeping the CPU and other components at suitable temperatures during long working sessions, this would allow for smoother performance, and the case is compatible to fit all the components inside with a lot of room for air flow.

Windows 11 Pro:

I chose Windows 11 Pro as it is the industry standard because of its compatibility with all the applications needed for project management, it can run the project managers' business software, spreadsheets, and communications, and provides a professional and widely recognized workplace.

Monitors, 2 × Pixio PX246 24-inch FHD IPS

I chose two 24-inch monitors because the assignment specifically requires a strong display workflow. The dual screen setup allows the Project Manager to view timelines, spreadsheets, emails, documents, and project information side-by-side without constantly switching between windows, this would help with keeping everything organized and reduce the time that is needed for the task without constant switching.

Keyboard + Mouse Dell KM5221W

I chose the Dell keyboard and mouse because they provide the basic input devices required for everyday office work. They allow the Project Manager to efficiently type reports, update spreadsheets, manage documents and navigate applications; they are reasonably priced and are dependable for everyday tasks.

Webcam, NexiGo N60 1080p

The NexiGo N60 is suitable for the project manager as it is high quality and reliable webcam that the Project manager needs for regular face to face calls with clients, where the web quality needs to be clean and reliable, the 1080p is a sufficient resolution that all devices are able to display, abs is the standard. This webcam provides suitable video quality for professional online meetings.

Headset – Sades SA-902-B:

I chose the Sades headset because clear audio is important for client meetings, team communication, and online calls; clear and precise audio quality is important for clear communication. The headset also provides a microphone for two-way communication while helping the Project Manager hear meetings clearly. The microphone's quality is dependable and good as well, which is needed for clear communication.

Architecture / CAD Designer:

1. Role Description:

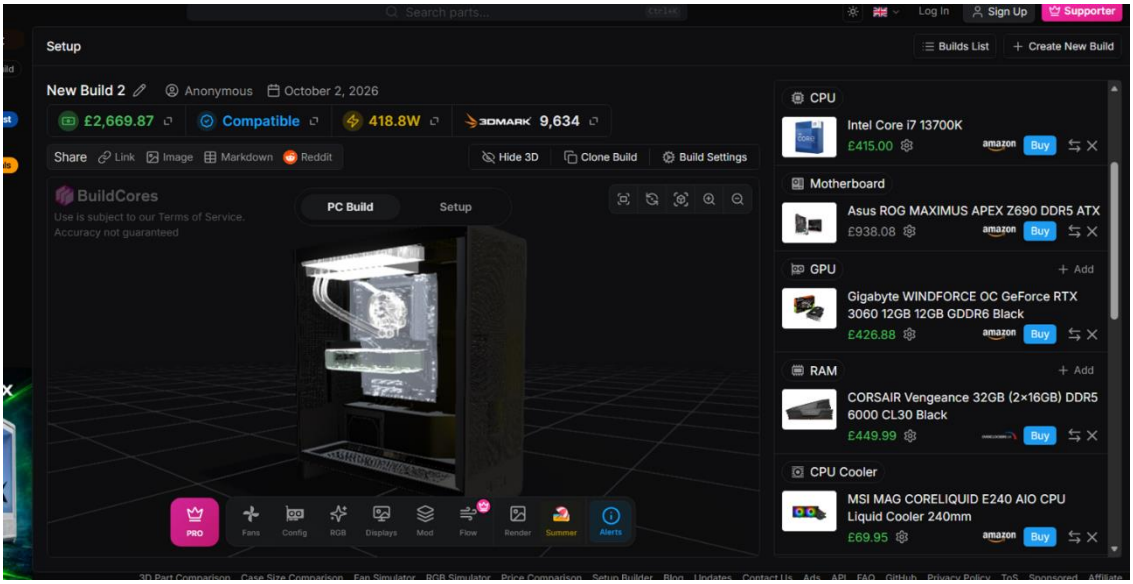
The Architect / CAD Designer role requires a workstation capable of handling:

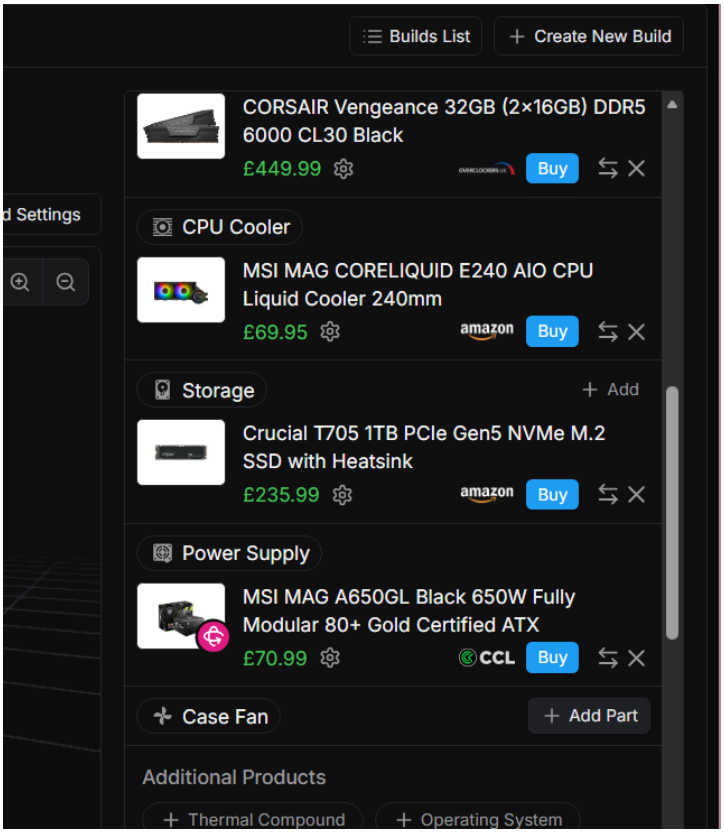
- High precision 2D CAD drawings
- Complex 3D architectural models
- Large BIM datasets
- Multiapplication workflows (AutoCAD, Revit, ArchiCAD, SketchUp, Rhino)
- High-resolution visual output for client presentations

This role demands strong CPU performance, dedicated GPU acceleration, fast NVMe storage, and colour accurate displays. It remains responsive even when working with large building models, detailed floor plans, and multiple linked files.

2. Base Unit Specification (Build Cores):

Build Cores screenshot shows a fully compatible Intel CAD workstation with excellent performance for architectural workloads.





CPU — Intel Core i7-13700K

A high-performance processor with exceptional single core speed, ideal for CAD operations such as regenerating views, constraint solving, and geometry calculations.

Motherboard — ASUS ROG Maximus Apex Z690 (DDR5)

A premium ATX motherboard supporting DDR5 memory, PCIe 4.0, and excellent stability for professional workloads.

RAM — 32GB DDR5 (Corsair Vengeance)

Perfect for handling large BIM files, multilayer CAD drawings, and multiple applications running simultaneously.

Storage — Crucial T705 1TB NVMe SSD (PCIe Gen5)

Ultrafast NVMe storage ensures rapid loading of project files, textures, and linked models.

GPU — Gigabyte RTX 3060 12GB

A dedicated GPU required for 3D modelling, viewport rendering, and real-time visualisation previews.

CPU Cooler — MSI MAG CORELIQUID E240 (240mm AIO)

Keeps the Intel CPU cool during long CAD sessions, preventing thermal throttling.

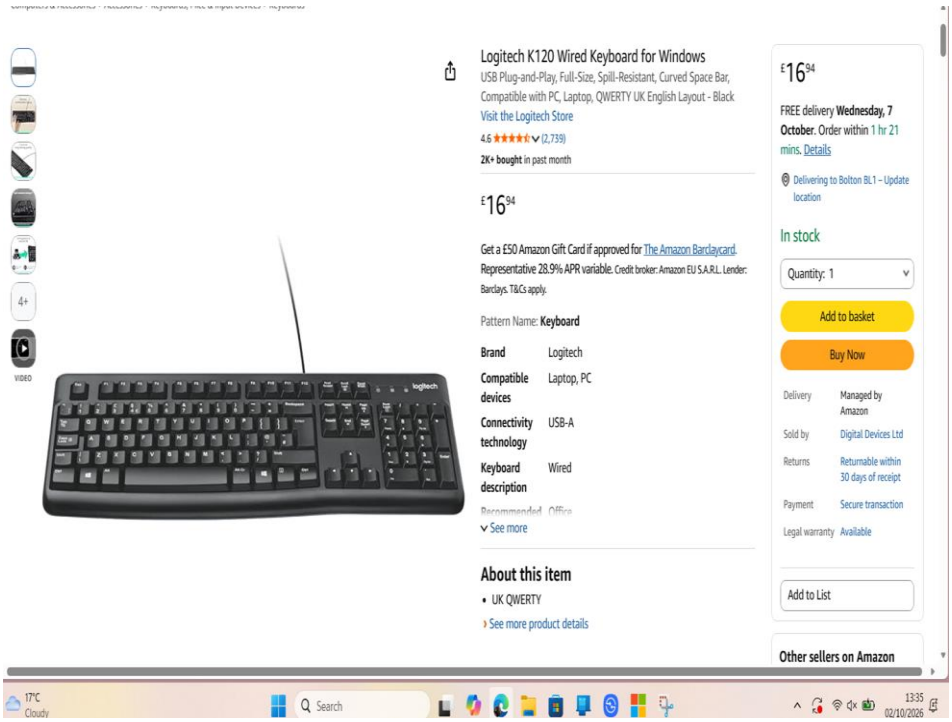
PSU — MSI MAG A650GL 650W Gold

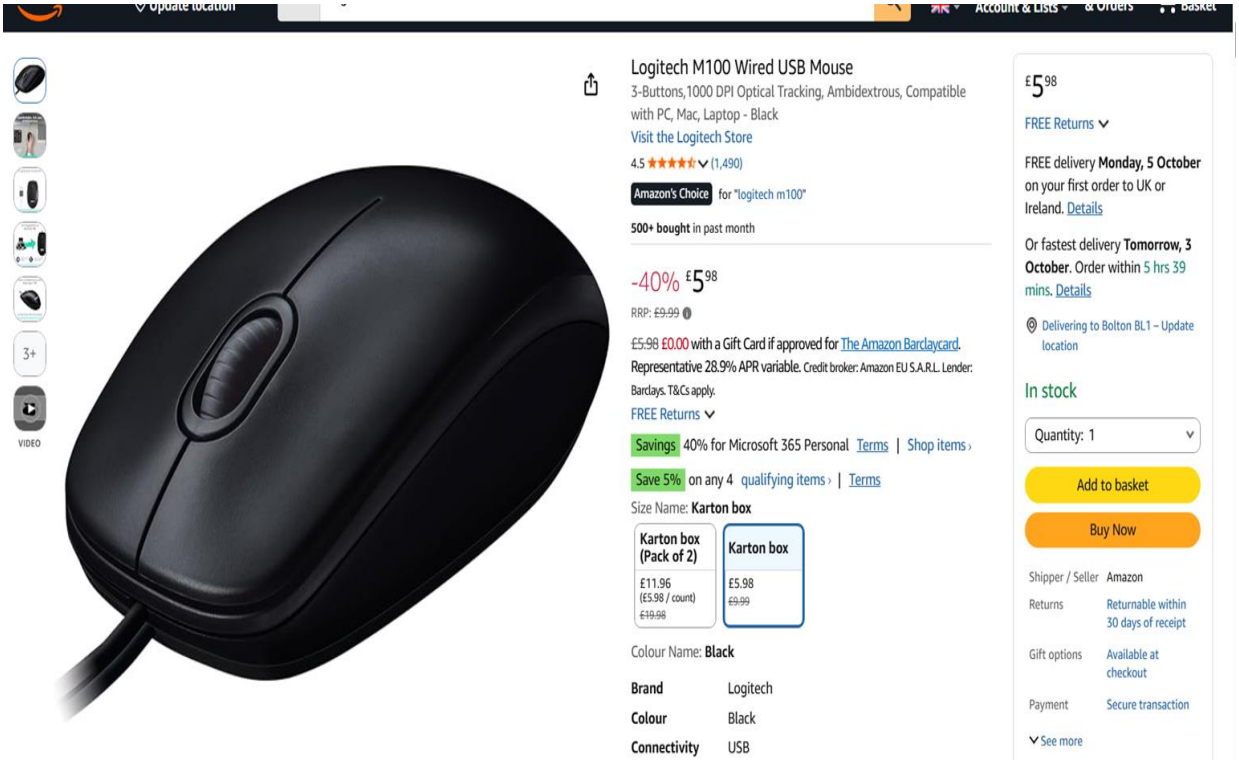
Reliable, efficient power delivery suitable for GPU-heavy workloads.

Peripheral Specification (Amazon UK Screenshots)

Amazon screenshots show all required peripherals for the CAD workstation.







Logitech M100 Wired USB Mouse

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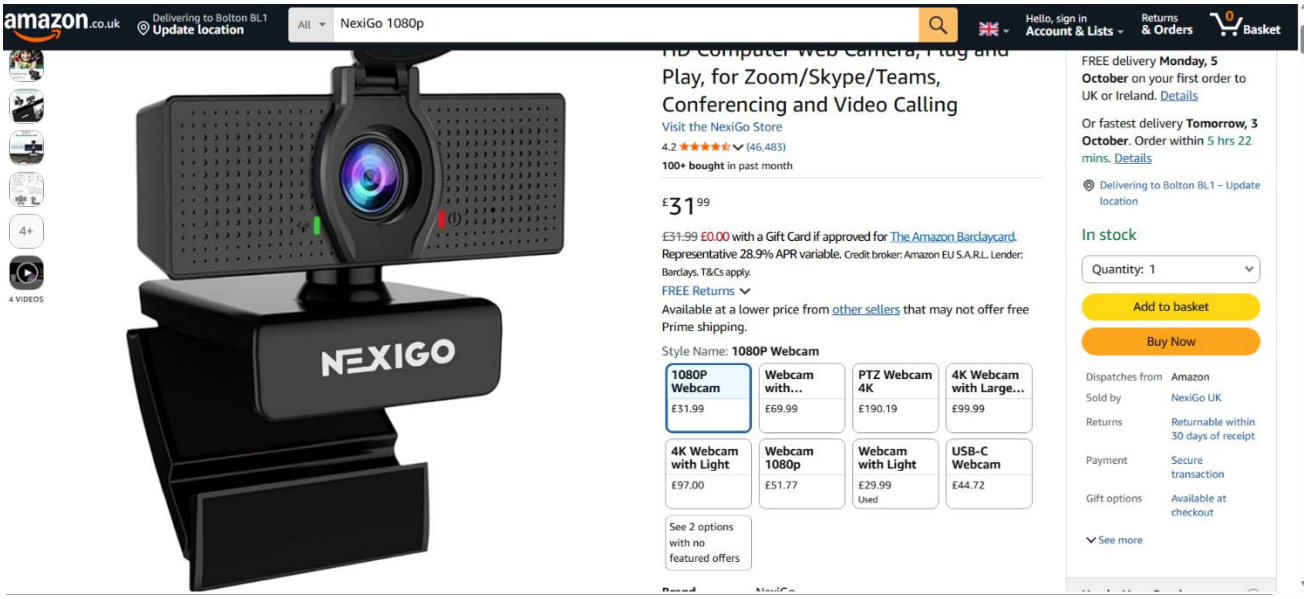
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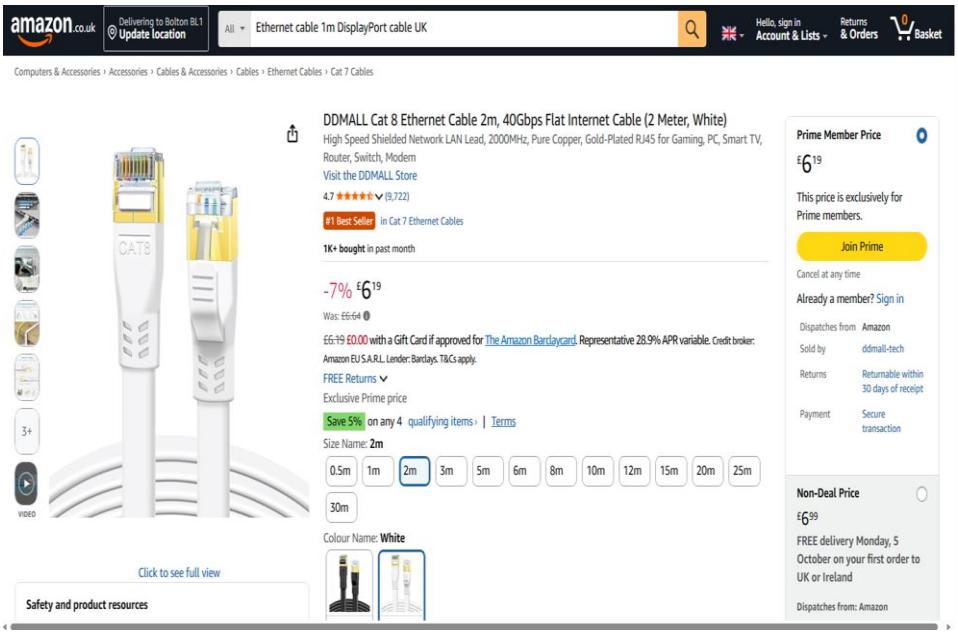
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Colour Name: White

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Safety and product resources

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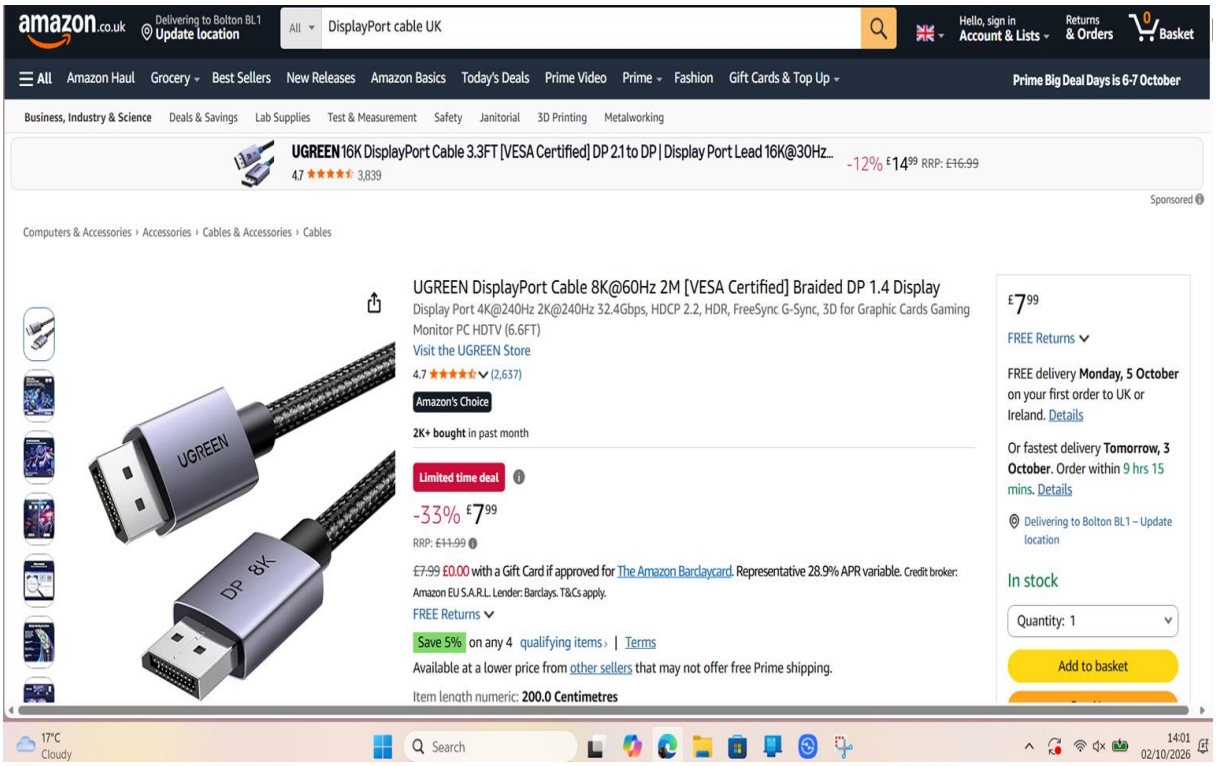
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Dispatches from: Amazon



Monitor — LG Ultra Gear 25" IPS (1080p, 200Hz)

Although gaming focused, the IPS panel provides excellent colour accuracy and clarity for CAD drawings.

Keyboard — Logitech K120 Wired Keyboard

A reliable full-size keyboard suitable for long drafting sessions.

Mouse — Logitech M100 Wired Mouse

Smooth, precise tracking ideal for CAD operations.

Webcam — NexiGo 1080p HD Webcam

Perfect for remote client meetings and design reviews.

DisplayPort Cable — UGREEN 8K DP Cable

Ensures stable high-resolution output to the monitor.

Ethernet Cable — DDmall Cat 8 Cable

Provides fast, stable network connectivity for cloud-based BIM workflows.

3. Updated Cost Table:

Category	Item	Price	Source
CPU	Intel Core i7-13700K	£415.00	BuildCores

Motherboard	ASUS ROG Maximus Apex Z690	£938.08	BuildCores
RAM	Corsair Vengeance 32GB DDR5	£449.99	BuildCores
Storage	Crucial T705 1TB NVMe SSD	£235.99	BuildCores
GPU	Gigabyte RTX 3060 12GB	£426.88	BuildCores
CPU Cooler	MSI MAG CORELIQUID E240	£69.95	BuildCores
PSU	MSI MAG A650GL 650W Gold	£70.99	BuildCores
Monitor	LG Ultra Gear 25" IPS	£99.95	Amazon
Keyboard	Logitech K120	£16.94	Amazon
Mouse	Logitech M100	£5.98	Amazon
Webcam	NexiGo 1080p HD	£31.99	Amazon
DisplayPort Cable	UGREEN 8K DP Cable	£7.99	Amazon
Ethernet Cable	DDmall Cat 8 Cable	£6.19	Amazon

Total Workstation Cost:

£2,775.82

Budget Per Workstation:

£2,500

Over Budget:

£275.82

4. Component Justification:

CPU — Intel Core i7-13700K

Provides exceptional single core performance, which is critical for CAD tasks such as regenerating views, updating geometry, and processing constraints.

GPU — RTX 3060 12GB

Handles 3D modelling, viewport rendering, and real-time visualisation smoothly.

RAM — 32GB DDR5

Ensures stable performance when working with large BIM files and multiple applications.

Storage — NVMe Gen5 SSD

Ultrafast loading and saving of project files improve workflow efficiency.

Monitor — IPS Panel

IPS technology provides accurate colours and wide viewing angles, essential for design work.

5. Ethical & Professional Considerations

This workstation design follows responsible purchasing principles:

- **Avoids overspecification -** no unnecessary high-end components
- **Energy efficient hardware -** reduces long-term running costs
- **Durable components -** ensure longevity and reduce e-waste
- **Secure storage** protects sensitive architectural files
- **Role appropriate design -** ensures staff productivity without wasting company funds
- **Supports accessibility -** through clear displays and ergonomic peripherals

This aligns with professional IT procurement standards and ethical handling of client information.

3D visualisation Specialist:

1. Role Description

A 3D Visualisation Specialist is responsible for creating high-quality visual representations of architectural projects. This includes producing detailed 3D models, rendering realistic images, applying textures and materials, and creating visual presentations for clients and stakeholders.

The role frequently works with large project files, complex scenes, high-resolution textures, and rendering software such as Blender. These workloads require a powerful workstation capable of processing demanding graphical and computational tasks efficiently.

To support these requirements, the workstation must provide strong CPU performance, professional graphics capabilities, large memory capacity, fast storage, and a high-resolution display for accurate visualisation work.

Typical tasks

- Creating and editing 3D models and scenes
- Producing high-quality visualisations and renders
- Working with complex graphical projects
- Manipulating large 3D assets and textures
- Viewing and editing detailed scenes in real time
- Rendering finished visualisations

2. Hardware requirements

The workstation requires:

- A high-performance **CPU** for rendering and scene calculations.
- A powerful dedicated **GPU** for 3D viewport performance and GPU-accelerated workloads
- Sufficient **RAM** capacity (64 GB) for large and complex scenes, multitasking, and rendering workloads.
- Fast and sufficiently large **SSD storage** for applications, projects, textures, and assets
- A suitable **motherboard** that supports the selected components and future upgrades.
- An appropriate **power supply** to safely power the high-performance components
- Effective **CPU cooling** and case airflow.
- A high-quality **monitor** suitable for detailed visual work and model inspection.
- A reliable **keyboard** and **mouse** peripherals for day-to-day workstation use.

3. Build Specification

Component	Product	Cost
Case	Montech HS02 ATX Mid Tower	**£24.98**
CPU	AMD Ryzen 9 7900X	**£253.00**
Motherboard	ASUS TUF GAMING B850-PLUS WIFI	**£159.99**

GPU	Spark Intel Arc Pro B60 24GB	**£869.99**
RAM	Corsair Vengeance RGB 64GB DDR5 6000	**£976.99**
CPU Cooler	ID-COOLING FX240 PRO 240mm	**£33.04**
Storage	WD Green SN350 2TB NVMe SSD	**£213.74**
PSU	MSI MAG A850GL 850W Gold	**£99.99**
Monitor	ASUS ROG Swift OLED PG32UCDP 4K	**£749.00**
Keyboard	Redragon K745	**£43.99**
Mouse	Logitech M220 Wireless	**£11.99**

Total Per Workstation **£3,436.70**

4. Cost Table

Item	Cost	Justification
CPU	**£253.00**	Multi-core processor suitable for rendering and modelling workloads.
GPU	**£869.99**	Professional 24GB graphics card for complex 3D scenes and visualisation.
RAM	**£976.99**	64GB supports large projects, multitasking, and rendering workloads.
Storage	**£213.74**	2TB capacity suitable for large assets and project files.
Monitor	**£749.00**	4K display enables detailed viewing of models and renders.
Remaining Components	**£373.98**	Supporting hardware required for stability and usability.

Budget Calculation

Per Workstation Budget: £3,750

Actual Workstation Cost: £3,436.70

Difference: **£313.30 Under Budget**

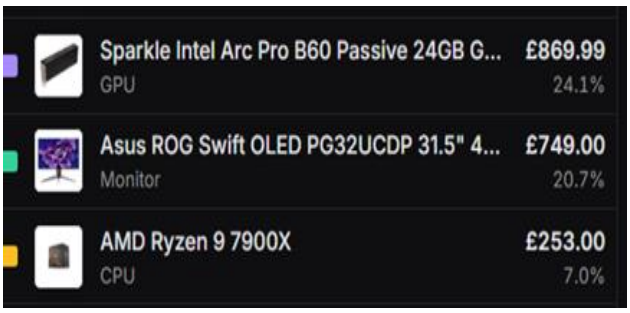
Role Quantity: 2 Workstations

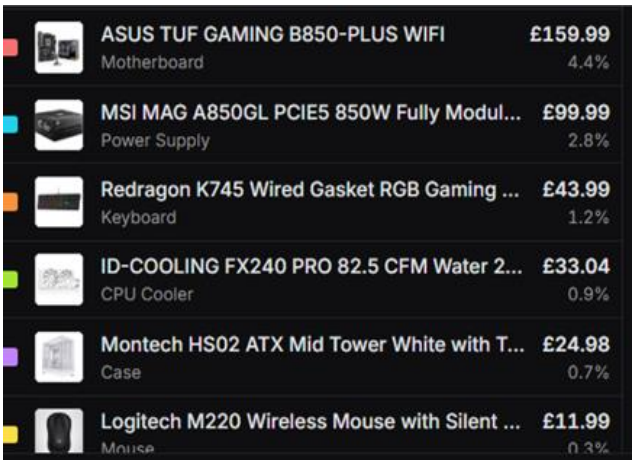
Total Role Cost: **£6873.40**

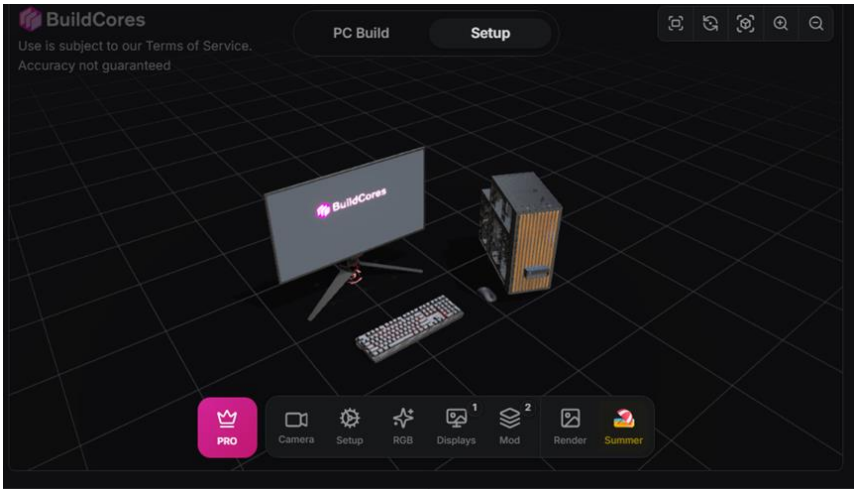
Role Allocation: **£7,500**

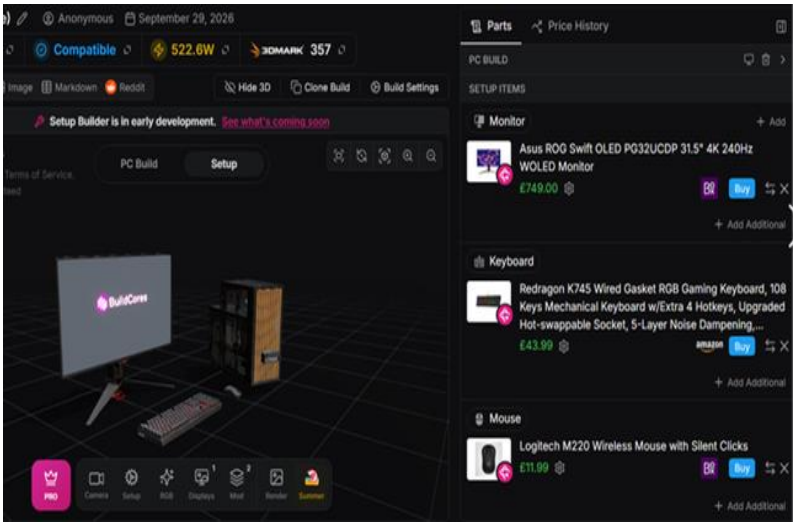
Remaining: **£626.60 Under Budget**

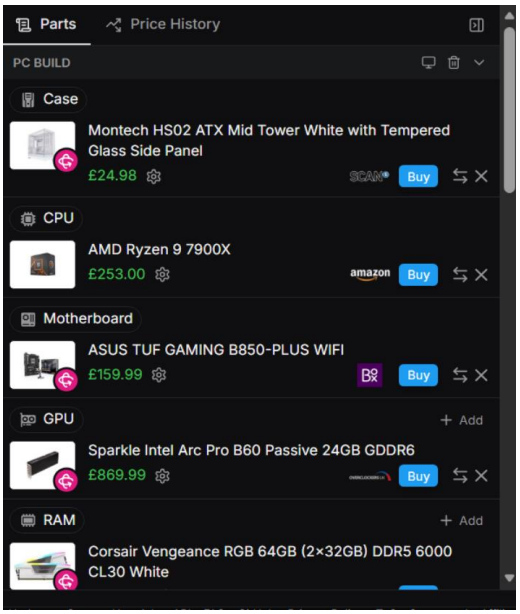
5. Supporting screenshots

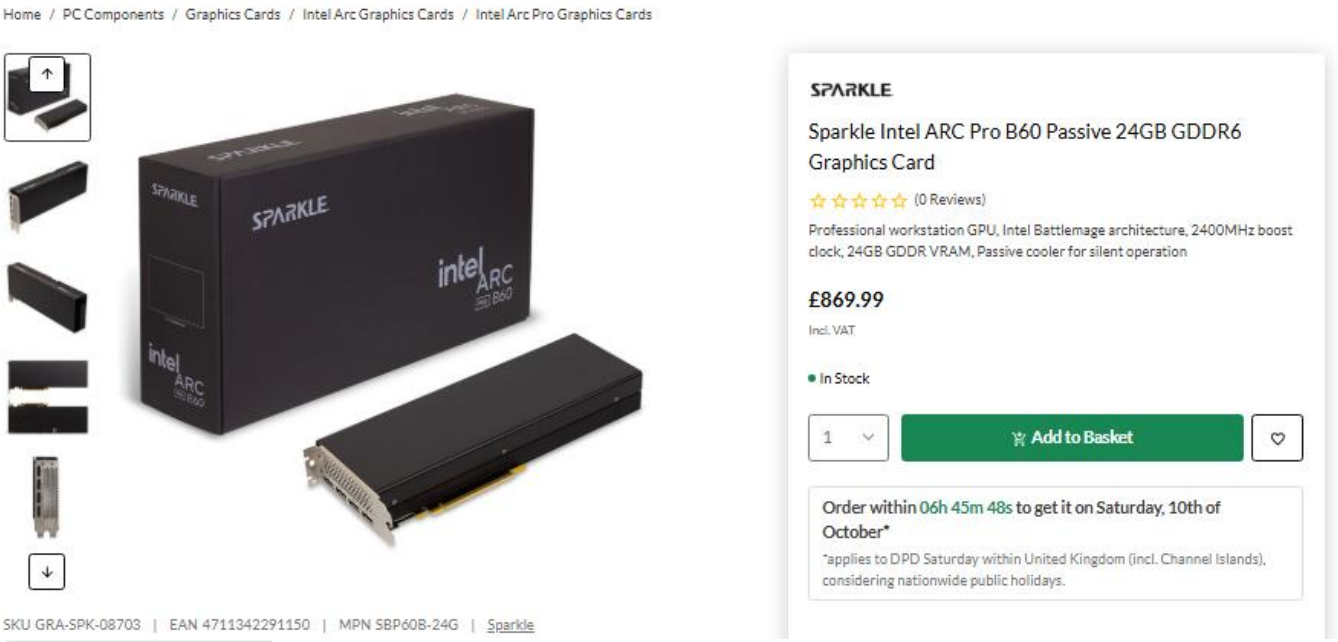


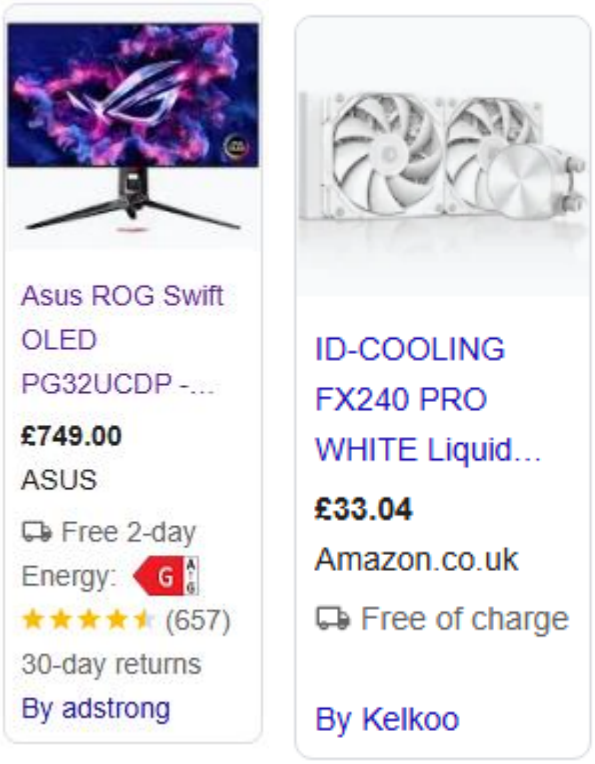












6. Component Justification

CPU

The AMD Ryzen 9 7900X was selected because its 12 cores provide strong multi-threaded performance for rendering, modelling, and scene processing. Compared with the previously considered Ryzen 9 9950X3D, it offers significant cost savings while still meeting the performance requirements of a 3D Visualisation Specialist.

GPU

The chosen GPU was selected because its 24GB VRAM capacity supports complex 3D scenes, high-resolution textures, and GPU-accelerated rendering workloads. The large memory capacity makes it suitable for professional visualisation projects and Blender based workflows.



RAM

The Corsair Vengeance RGB 64GB DDR5 memory kit provides sufficient capacity for large project files, multitasking, rendering, and complex scene management. The 64GB configuration helps ensure smooth performance when working with demanding visualisation workloads.

Storage

The WD Green SN350 2TB NVMe SSD provides fast access to applications, project files, and large graphical assets. The 2TB capacity allows storage of software, active projects, and rendered output files without requiring additional drives.

Monitor

The selected monitor was selected because of its 4K resolution which allows detailed viewing of 3D models, textures, and rendered images. Accurate visual representation is important when producing professional architectural visualisations.

PSU

The MSI MAG A850GL 850W power supply provides sufficient power for all workstation components while maintaining reliability and allowing future upgrade potential.

Cooling

The ID-COOLING FX240 PRO liquid cooler helps maintain stable processor temperatures during demanding workloads such as rendering and extended modelling sessions.

Keyboard and mouse

The selected keyboard and mouse provide reliable input devices for everyday workstation use without unnecessarily increasing the overall cost of the build.

7. Compromises / Trade-offs

The workstation budget required a balance between performance and cost. The original configuration included an AMD Ryzen 9 9950X3D processor. However, this was replaced with the AMD Ryzen 9 7900X to reduce overall cost while maintaining strong rendering and visualisation performance.

Premium aesthetic features and unnecessary peripherals were avoided so that more of the budget could be allocated to core workstation components such as the GPU, memory, and display.

8. Ethical / Responsible Purchasing

The recommended workstation provides opportunities for future upgrades through the AMD AM5 platform helping to extend its useful lifespan and reduce electronic waste. Components should be sourced from reputable suppliers and selected with consideration for energy efficiency, warranty support, and responsible recycling at the end of their service life.

Final Budget calculations:

Per-Workstation Unit Cost Breakdown

Staff Employee Role	Budget Per Unit	Actual Cost Per Unit	Unit Variance	Unit Budget Status
Administrative Assistant	£950.00	£657.03	-£292.97	**£292.97 Under Budget**
Project Manager	£1,300.00	£1,226.94	-£73.06	**£73.06 Under Budget**
Architect / CAD Designer	£2,500.00	£2,775.82	+£275.82	**£276.02 Over Budget**
3D Visualisation Specialist	£3,750.00	£3,436.70	-£313.30	**£313.30 Under Budget**

Total Studio Procurement Summary

Staff Employee Role	Workstation Qty	Total Allocated Budget	Total Actual Cost	Total Role Variance	Final Role Status
Administrative Assistant	3	£2,850.00	£1,971.09	-£878.91	**£878.91 Under Budget**
Project Manager	2	£2,600.00	£2,453.88	-£146.12	**£146.12 Under Budget**
Architect / CAD Designer	5	£12,500.00	£13,879.10	+£1,379.10	**£1,379.10 Over Budget**
3D Visualisation Specialist	2	£7,500.00	£6,873.40	-£626.60	**£626.60 Under Budget**
STUDIO TOTALS	**12**	**£25,450.00**	**£25,178.47**	**- £272.53**	**£272.53 Under Budget**

Overall Summary:

In conclusion, this procurement plan successfully gives Northwest Architecture Studio Ltd the exact hardware needed for all 12 staff workstations. The base cost for all hardware comes to £25,177.47, which runs £272.53 under the base hardware budget of £25,450.00. This success was driven by a smart balancing strategy, where the savings carved out from the Administrative Assistant, Project Manager, and 3D Visualisation machines easily absorbed the extra cost needed to buy powerful, dedicated graphics cards for the Architect / CAD Designer tier. Furthermore, because we did not touch the £2,000.00 contingency fund, the overall project completes a massive £2,272.53 under the maximum company limit.

To make sure all the math and parts were 100% correct, our team built an SQL database to check every single price and component connection. Running these database tests proved that our calculations are completely accurate and free of errors. The final list provides the company with a safe, affordable, and high-performing computer setup that perfectly fits what the business asked for.